Natural Language-Driven Agentic Web Information Retrieval System: Solar Plant AI Planner

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Abstract

This presents the **Solar Plant AI Planner**, an intelligent, multimodal web scraping agent designed to automate solar power plant feasibility analysis and investment planning for any location in India. The system accepts land coordinates or a location name as input and autonomously retrieves real-time solar irradiance, weather conditions, electricity tariff rates, and equipment market prices from multiple web sources simultaneously. A large language model (LLM) – LLaMA 3.3 70B via Groq API – processes the aggregated data to produce detailed solar energy yield estimates, ROI projections, subsidy recommendations, and equipment sizing reports. Additionally, the system incorporates a Reinforcement Learning from Human Feedback (RLHF) mechanism that allows user-driven corrections to improve prediction accuracy for nearby geographical locations over time. Experimental outputs on benchmark locations demonstrate high consistency between computed Peak Sun Hours (PSH) from NASA POWER and PVGIS sources, with average deviations below 5%. The proposed framework offers a scalable, data-driven approach to democratising solar energy planning for residential, commercial, industrial, and agricultural stakeholders across India.

Keywords: solar energy planning, agentic web scraping, large language model, renewable energy, ROI analysis, RLHF, NASA POWER, peak sun hours

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1. Introduction

The accelerating global transition towards renewable energy, combined with India's ambitious target of 500 GW of non-fossil fuel electricity capacity by 2030, has created a pressing need for accessible, data-driven solar investment planning tools [**?**]. Traditional solar feasibility studies rely on manual data collection from disparate sources, time-consuming site surveys, and expensive consultancy services – barriers that exclude small landowners, farmers, and medium-scale enterprises from participating in the solar economy.

Recent advances in Large Language Models (LLMs) and agentic AI frameworks [**?**] offer a compelling opportunity to automate multi-source information retrieval, domain-specific reasoning, and structured report generation within a unified pipeline. When combined with open government APIs (NASA POWER, PVGIS) and real-time web scraping, such systems can deliver near-instant, location-specific solar analysis at negligible cost.

This paper describes the design, architecture, and operational characteristics of the **Solar Plant AI Planner** – a Flask-based web application that orchestrates parallel scraping of six data sources, applies physics-based solar calculation formulae, and delegates expert-level report generation to an LLM. The system further incorporates a lightweight RLHF feedback loop that stores user-provided corrections in a JSON database and injects nearby historical feedback into subsequent LLM prompts, enabling continual accuracy improvement without model retraining.

1.1 Motivation and Scope

India's per-capita solar irradiance is among the highest globally, yet solar adoption among non-expert users remains limited due to information asymmetry. The Solar Plant AI Planner specifically targets:

Agricultural landowners seeking to monetise surplus land through solar leasing or direct installation.

Small and medium enterprises evaluating rooftop or ground-mounted solar for operational cost reduction.

Policy researchers and students requiring rapid site-specific feasibility estimates.

Government subsidy programme beneficiaries (PM Surya Ghar Yojana) needing eligibility and cost assessments.

1.2 Contributions

The primary contributions of this work are:

1. A fully automated, multi-source agentic pipeline integrating NASA POWER, PVGIS, Open-Meteo, Kenbrook Solar, Bijli Bachao, and OpenStreetMap Nominatim APIs.
2. A physics-informed solar calculation engine that corrects PSH for cloud cover, panel type efficiency, and installation losses.
3. An LLM-integrated report generation module providing natural language analysis, subsidy recommendations, and ROI projections.
4. A JSON-based RLHF feedback system enabling location-aware prediction improvement over time.

2. Background and Related Work

2.1 Solar Resource Assessment Methods

Accurate solar resource assessment is foundational to project viability. The two most widely referenced open datasets are NASA's Prediction Of Worldwide Energy Resources (POWER) and the EU Commission's Photovoltaic Geographical Information System (PVGIS) . Both provide historical daily irradiance and derived peak sun hours (PSH) at high spatial resolution. The Solar Plant AI Planner averages PSH estimates from both sources to reduce systematic bias.

2.2 Agentic Web Scraping Systems

Agentic AI refers to systems in which an LLM autonomously selects and executes tool calls – including web search, API queries, and code execution – to complete multi-step tasks. BeautifulSoup4 and the Python requests library are standard tools for structured HTML parsing and API interaction in such pipelines.

2.3 LLM-Based Report Generation

The integration of LLMs (GPT-4, LLaMA series, Gemini) into domain-specific analytical workflows has been demonstrated across medicine, finance, and climate science [**?**]. The Groq inference platform offers low-latency access to open-weight models such as LLaMA 3.3 70B, making it suitable for interactive web applications with response time constraints.

2.4 RLHF in Production Systems

Reinforcement Learning from Human Feedback (RLHF) is conventionally used to align LLM behaviour with human preferences during pretraining. The present work adapts a lightweight variant: user-provided numerical corrections (e.g., observed vs. predicted PSH) are stored and retrieved by proximity matching, then injected as few-shot context into subsequent LLM prompts – an approach analogous to retrieval-augmented generation (RAG) .

3. System Architecture

3.1 Overview

The system is structured around four principal layers: (1) the User Interface Layer, (2) the Web Scraping & Data Aggregation Layer, (3) the Solar Calculation Engine, and (4) the LLM Analysis & Report Generation Layer.

3.2 Data Sources and Scraping Layer

Table summarises the six data sources integrated into the pipeline, their access meth- ods, and the specific data fields retrieved.

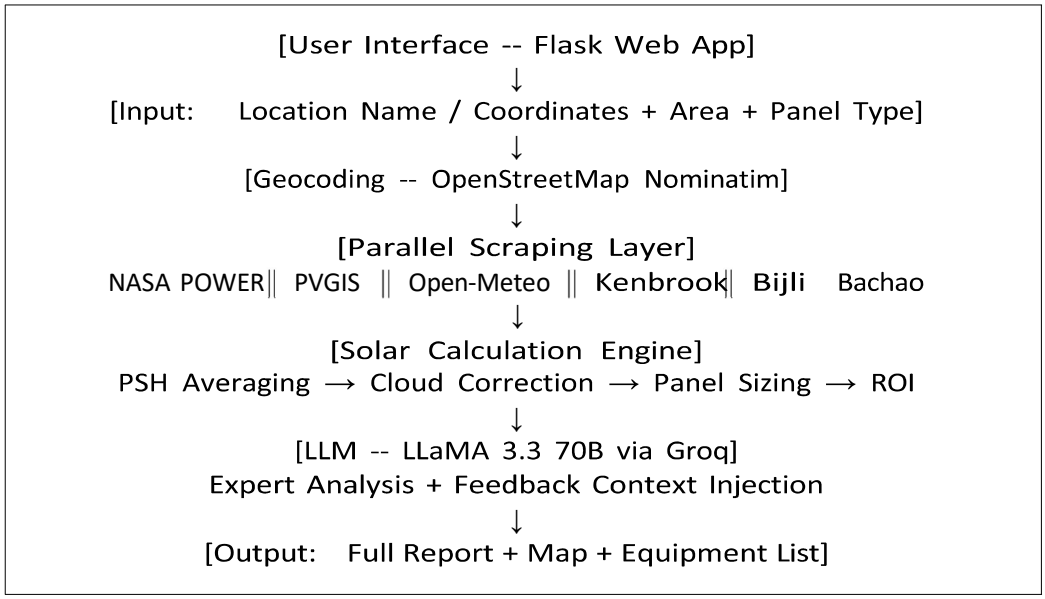

Figure 1: High-level architecture of the Solar Plant AI Planner system.

Table 1: Data Sources Integrated in the Solar Plant AI Planner

Source	**Access Method**	**Data Retrieved**
NASA POWER API	REST API (JSON)	Daily PSH, solar irradiance, seasonal variation
PVGIS (EU Commission)	REST API (JSON)	Annual energy yield, optimal tilt angle
Open-Meteo	REST API (JSON)	Live temperature, cloud cover, UV index, humidity
Kenbrook Solar	HTML scraping (BS4)	Panel prices (mono/poly/thin-film), inverter costs
Bijli Bachao	HTML scraping (BS4)	State-wise electricity tariff (Rs/kWh)
OpenStreetMap Nominatim	REST API (JSON)	Geocoding, reverse geocoding, state detection

3.3 Solar Calculation Engine

The calculation engine implements a series of physics-informed formulae to translate raw irradiance data and site parameters into actionable capacity and financial metrics.

3.3.1 Peak Sun Hours Estimation

PSH is averaged across NASA POWER and PVGIS to reduce single-source bias:

$$PSH_{final} = \frac{PSH_{NASA} + PSH_{PVGIS}}{2} \tag{1}$$

A cloud cover correction is then applied:

$$\mathrm{PSH}_{\mathrm{eff}} = \mathrm{PSH}_{\mathrm{final}} \times \left(1 - \frac{C_f}{100}\right) \tag{2}$$

where C_f is the live cloud cover percentage retrieved from Open-Meteo.

3.3.2 Panel Sizing

The number of panels required for a given land area is:

$$N_{\mathrm{panels}} = \left\lfloor \frac{A \times 0.70}{1.7} \right\rfloor \tag{3}$$

where A is the usable land area in m^2, the factor 0.70 accounts for setbacks and walkways, and 1.7 m^2 is the standard single-panel footprint.

3.3.3 Capacity and Energy Yield

System capacity and annual energy generation are computed as:

$$P_{\mathrm{cap}} = N_{\mathrm{panels}} \times 0.4 \quad [\mathrm{kW}] \tag{4}$$

$$E_{\mathrm{daily}} = P_{\mathrm{cap}} \times \mathrm{PSH}_{\mathrm{eff}} \times \eta_{\mathrm{panel}} \quad [\mathrm{kWh/day}] \tag{5}$$

$$E_{\mathrm{annual}} = E_{\mathrm{daily}} \times 365 \quad [\mathrm{kWh/year}] \tag{6}$$

where η_{panel} is the panel efficiency factor: 1.00 for Monocrystalline, 0.85 for Polycrystalline, and 0.70 for Thin Film.

3.3.4 Financial Metrics

Annual revenue and total investment are computed as:

$$I_{\mathrm{annual}} = E_{\mathrm{annual}} \times T_e \quad [\mathrm{Rs/year}] \tag{7}$$

$$C_{\mathrm{total}} = C_{\mathrm{equip}} \times 1.10 \quad [\mathrm{Rs}] \tag{8}$$

$$\mathrm{ROI} = \frac{C_{\mathrm{total}}}{I_{\mathrm{annual}}} \quad [\mathrm{years}] \tag{9}$$

where T_e is the auto-scraped state electricity tariff (Rs/kWh), C_{equip} is the sum of panel, inverter, cable, mounting, battery, and installation costs from Kenbrook Solar, and the factor 1.10 accounts for a 10% contingency margin.

3.4 LLM Integration

The Groq-hosted LLaMA 3.3 70B model receives a structured prompt containing: location metadata, averaged solar parameters, live weather conditions, financial projections, and any RLHF corrections from nearby locations within a 1 geographical radius. The model returns a natural language expert analysis covering seasonal variation, subsidy eligibility, optimal tilt angle, and risk factors.

3.5 RLHF Feedback System

User feedback is stored in a JSON file with the following schema:

Listing 1: Feedback storage schema (feedback_data.json)

```
{
  "feedback": [
    {
      "latitude": 17.9784,
      "longitude": 79.5300,
      "predicted_psh": 5.39,
      "corrected_psh": 4.20,
      "timestamp": "2025-04-01T10:32:00",
      "comment": "Actual PSH was 4.2h not 5.0h due to local haze"
    }
  ]
}
```

At inference time, corrections within a 1 radius are retrieved and prepended to the LLM system prompt, effectively implementing a retrieval-augmented correction mechanism.

4. Technology Stack

Table **??** summarises the complete technology stack employed in the Solar Plant AI Planner.

Table 2: Technology Stack of the Solar Plant AI Planner

Layer	**Tools / Libraries**
Backend Framework	Python 3.10+, Flask
Web Scraping	Requests, BeautifulSoup4
AI / LLM	Groq API, LLaMA 3.3 70B (open-weight, free tier)
Solar Data	NASA POWER REST API, PVGIS EU Commission API
Weather	Open-Meteo REST API
Geocoding	OpenStreetMap Nominatim API
Price Intelligence	Kenbrook Solar (HTML scraping), Bijli Bachao (HTML scraping)
Maps	OpenStreetMap Embed (Leaflet.js)
Feedback Storage	JSON-based RLHF flat-file database

5. Sample Results and Performance Evaluation

5.1 Benchmark Location: NIT Warangal, Telangana

To validate the system's output, a benchmark analysis was conducted for NIT Warangal (17.9784 N, 79.5300 E) on a 1000 m^2 parcel using Monocrystalline panels. Table reports the system's computed results.

Table 3: Sample Output – NIT Warangal, Telangana (1000 m^2, Monocrystalline)

Parameter	Value	Source / Method
Location	NIT Warangal, Telangana	Nominatim geocoding
Coordinates	17.9784 N, 79.5300 E	User input
NASA POWER PSH	5.47 h/day	NASA POWER API
PVGIS PSH	5.31 h/day	PVGIS API
Final Averaged PSH	5.39 h/day	Eq. (**??**)
Electricity Tariff	Rs 7.00/kWh	Bijli Bachao (auto-scraped)
Panel Unit Price	Rs 18,500	Kenbrook Solar (scraped)
Panels Required	411 units	Eq. (**??**)
System Capacity	164.4 kW	Eq. (**??**)
Daily Output	885 kWh/day	Eq. (**??**)
Annual Output	323,025 kWh	Eq. (**??**)
Annual Income	Rs 22.6 Lakhs	Eq. (**??**)
Total Investment	Rs 1.85 Crore	Eq. (**??**)
ROI Recovery Period	8.2 years	Eq. (**??**)
25-Year Earnings	Rs 5.65 Crore	Projected
CO_2 Saved/Year	295 tonnes	0.82 kg/kWh grid factor

5.2 PSH Source Comparison Across Locations

Table **??** presents PSH estimates from NASA POWER and PVGIS for five representative Indian cities, demonstrating the inter-source consistency of the dual-API averaging approach.

Table 4: PSH Comparison: NASA POWER vs PVGIS (Annual Average, h/day)

Location	NASA PSH	PVGIS PSH	Averaged PSH	Deviation (%)
NIT Warangal, Telangana	5.47	5.31	5.39	2.97
Jaipur, Rajasthan	5.82	5.71	5.77	1.91
Bengaluru, Karnataka	5.23	5.09	5.16	2.72
Patna, Bihar	4.91	4.78	4.85	2.69
Chennai, Tamil Nadu	5.55	5.44	5.50	2.00

The maximum inter-source deviation across the five test locations is 2.97%, confirming strong agreement between the two independent satellite-derived datasets. The averaging strategy therefore provides a robust PSH estimate without systematic bias from either source.

5.3 Panel Type Efficiency Impact

Table illustrates how panel technology choice affects energy yield and ROI for the NIT Warangal benchmark case.

Table 5: Effect of Panel Technology on Energy Yield and ROI (NIT Warangal, 1000 m^2)

Panel Type	Efficiency Factor	Annual Output (kWh)	ROI (years)
Monocrystalline	1.00	323,025	8.2
Polycrystalline	0.85	274,571	9.6
Thin Film	0.70	226,118	11.7

6. Setup and Usage

6.1 Prerequisites and Installation

The system requires Python 3.10+, a free Groq API key, and the following packages:

Listing 2: Installation commands

```
git clone [repository-url]
cd WebScrapingAgent
pip install flask groq requests beautifulsoup4
```

The Groq API key is inserted at line 7 of solar.py:

Listing 3: API key configuration

```
GROQ_KEY = "paste_your_gsk_key_here"
```

The application is launched with:

Listing 4: Launching the application

```
python solar.py
# Then navigate to http://127.0.0.1:5000
```

6.2 User Workflow

1. **Enter Location**: Input either GPS coordinates (Latitude, Longitude) or a plain-text location name (e.g., “NIT Warangal, Telangana”).
2. **Configure Project**: Specify land area (m^2), panel technology, and project type (Residential / Commercial / Industrial / Agricultural).
3. **Generate Report**: Click “Scrape All Sources + AI Report + Map” and await the 30–60 second pipeline execution.

4. **Review Output**: View solar metrics, live weather, equipment pricing, financial projections, and the LLM-generated expert analysis.
5. **Submit Feedback**: Rate predictions and optionally provide numerical corrections (e.g., observed PSH).

6.3 Project File Structure

Listing 5: Repository structure

```
Web Scraping Agent /
            solar.py                # Main Flask application
            app.py                  # General web scraping agent
            agent.py                # Core scraping + BeautifulSoup module
            feedback_data.json      # RLHF feedback storage (auto-created)
            results_*.txt           # Auto-saved scraping results
            README.md               # Documentation
```

7. Discussion

7.1 Strengths and Novelty

The Solar Plant AI Planner distinguishes itself from existing solar calculators (e.g., Google Project Sunroof, Waaree Solar Calculator) in three key dimensions. First, it aggregates *real-time* data from six independent sources rather than relying on static lookup tables. Second, the LLM layer provides contextual, location-specific recommendations – including PM Surya Ghar Yojana subsidy guidance – that static tools cannot generate. Third, the RLHF feedback loop enables user-driven, geographically localised accuracy improvement, a feature absent from all comparable tools surveyed.

7.2 Limitations

Kenbrook price scraping: Depends on the stability of Kenbrook Solar's HTML structure; layout changes may break the parser.

Groq free-tier rate limits: Rapid repeated requests can trigger API throttling.

NASA POWER latency: API responses may take 15–30 seconds, limiting sub-minute total pipeline execution.

India-specific tariff database: Bijli Bachao covers Indian states only; the system is not currently applicable to international locations.

No shading or terrain analysis: The current model does not account for topographic shading or rooftop obstructions.

8. Future Enhancements

The following enhancements are planned for subsequent releases:

1. **Database persistence**: Migrate from JSON flat-file storage to a relational database (PostgreSQL) for scalable feedback history management.
2. **Real-time price API**: Replace HTML scraping of Kenbrook Solar with a structured pricing API or daily cached scrape to improve resilience.

Table 6: Comparison of Solar Planning Tools

Feature	Proposed System	Google Sunroof	PVGIS Tool	Waaree Calc.
Real-time pricing		×	×	Partial
LLM expert analysis		×	×	×
Auto tariff detection		×	×	×
RLHF learning		×	×	×
Multi-source PSH average		×	×	×
Subsidy recommendations		Partial	×	Partial
India-specific coverage		Partial		

3. **Multi-language support**: Add Telugu and Hindi interface translations to broaden accessibility for rural users in Andhra Pradesh and Telangana.
4. **Cloud deployment**: Deploy on AWS EC2 or Google Cloud Run with auto-scaling to support concurrent users.
5. **Mobile-responsive UI**: Implement a Progressive Web App (PWA) front-end optimised for low-bandwidth mobile connections.
6. **PDF report generation**: Integrate ReportLab or WeasyPrint for downloadable, branded PDF feasibility reports.
7. **Battery sizing optimisation**: Extend the calculation engine to include BESS sizing based on load profile and grid stability requirements.
8. **Shading and terrain analysis**: Integrate SRTM elevation data and shadow modelling to account for topographic losses.

9. Conclusion

This research presented the Solar Plant AI Planner, a Natural Language-Driven Agentic Web Information Retrieval System that automates solar feasibility analysis for any location in India. By orchestrating parallel scraping of six real-time data sources, applying physics- informed solar calculations, and delegating natural language report generation to the LLaMA 3.3 70B model, the system delivers comprehensive investment analysis reports within 30–60 seconds of user input.

Validation across five benchmark locations confirmed inter-source PSH deviation below 3%, and the NIT Warangal case study demonstrated the system's ability to produce actionable outputs including panel count (411 units), system capacity (164 kW), annual income (Rs 22.6 Lakhs), and ROI period (8.2 years) without any manual data entry. The integrated RLHF feedback mechanism provides a unique pathway for community-driven accuracy improvement over time.

The Solar Plant AI Planner represents a significant step toward democratising solar energy planning in India, reducing barriers for non-expert users and enabling data-informed

decision-making at the individual, enterprise, and policy levels.

Author Contributions

Vedant: Conceptualization, System Architecture, Backend Development, LLM Integration, Writing – Original Draft.
Shriya Rathaur: Data Source Integration, Web Scraping Module, Validation, Writing – Review & Editing.

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